

Table 1: Correction payloads, paths, and tags used in the analysis.

Era	Correction	Payload / path	Key / tag
Run3_2022	JEC/JER	/cvmfs/cms-griddata.cern.ch/cat/metadata/JME/Run3-22CDSep23-Summer22-NanoA0Dv12/latest/jet_jerc.json.gz	JEC MC: Summer22_22Sep2023_V4_MC; JEC data: Summer22_22Sep2023_V4_DATA; JER: Summer22_22Sep2023_JRV2_MC
	JER smearing	/cvmfs/cms-griddata.cern.ch/cat/metadata/JME/JER-Smearing/latest/jer_smear.json.gz	sources: JER, Total
	Pileup	/cvmfs/cms-griddata.cern.ch/cat/metadata/LUM/Run3-22CDSep23-Summer22-NanoA0Dv12/latest/puWeights.json.gz	Collisions2022_355100_357900_eraBCD_GoldenJson
	Muon SF	/cvmfs/cms-griddata.cern.ch/cat/metadata/MUO/Run3-22CDSep23-Summer22-NanoA0Dv12/latest/muon_Z.json.gz	NUM_LooseID_DEN_TrackerMuons, NUM_MediumID_DEN_TrackerMuons, NUM_TightID_DEN_TrackerMuons, NUM_LoosePFIso_DEN_LooseID, NUM_LoosePFIso_DEN_MediumID, NUM_LoosePFIso_DEN_TightID, NUM_TightPFIso_DEN_TightID, NUM_LooseMiniIso_DEN_LooseID, NUM_LooseMiniIso_DEN_MediumID, NUM_IsoMu24_DEN_CutBasedIdMedium_and_PFIsoMedium, NUM_IsoMu24_DEN_CutBasedIdTight_and_PFIsoTight
	Muon ScaRe	corrections/data/MUO/MuonScaRe/Run3-22CDSep23-Summer22-NanoA0Dv12/muon_scalesmearing.json	standard; VXBS payload: corrections/data/MUO/MuonScaRe/Run3-22CDSep23-Summer22-NanoA0Dv12/muon_scalesmearing_VXBS.json
	Muon FSR	corrections/muon_fsr.py	algorithmic; no payload
	Jet veto map	/cvmfs/cms-griddata.cern.ch/cat/metadata/JME/Run3-22CDSep23-Summer22-NanoA0Dv12/latest/jetvetomaps.json.gz	Summer22_23Sep2023_RunCD_V1
	b-tag WP	/cvmfs/cms-griddata.cern.ch/cat/metadata/BTV/Run3-22CDSep23-Summer22-NanoA0Dv12/latest/btagging.json.gz	PNet_wp_values (L, M, T)
	Golden JSON	corrections/data/golden_json/Cert_Collisions2022_355100_362760_Golden.json	data luminosity mask
	Trigger SF	HLT_IsoMu24	NUM_IsoMu24_DEN_CutBasedIdMedium_and_PFIsoMedium
Run3_2022EE	LHE PDF uncertainty	LHEPdfWeight	replicas_rms: nominal index 0; replicas 1-100; up=1+RMS, down=max(0,1-RMS); alphaS excluded; missing branch=unity
	LHE QCD scales	LHEScaleWeight	muR0p5_muF0p5=index 0; muR0p5_muF1=index 1; muR1_muF0p5=index 3; muR1_muF2=index 5; muR2_muF1=index 7; muR2_muF2=index 8; dataset-normalized six-point prescription; QCD_ren_scale_{process}: muR0p5_muF1 / muR2_muF1; QCD_fac_scale_{process}: muR1_muF0p5 / muR1_muF2; denominator sums are stored in skim reports; missing sums=skip
	JEC/JER	/cvmfs/cms-griddata.cern.ch/cat/metadata/JME/Run3-22EFGSep23-Summer22EE-NanoA0Dv12/latest/jet_jerc.json.gz	JEC MC: Summer22EE_22Sep2023_V4_MC; JEC data: Summer22EE_22Sep2023_Run{}_V4_DATA, Summer22EE_22Sep2023_V4_DATA; JER: Summer22EE_22Sep2023_JRV2_MC
	JER smearing	/cvmfs/cms-griddata.cern.ch/cat/metadata/JME/JER-Smearing/latest/jer_smear.json.gz	sources: JER, Total
	Pileup	/cvmfs/cms-griddata.cern.ch/cat/metadata/LUM/Run3-22EFGSep23-Summer22EE-NanoA0Dv12/latest/puWeights.json.gz	Collisions2022_359022_362760_eraEFG_GoldenJson
	Muon SF	/cvmfs/cms-griddata.cern.ch/cat/metadata/MUO/Run3-22EFGSep23-Summer22EE-NanoA0Dv12/latest/muon_Z.json.gz	NUM_LooseID_DEN_TrackerMuons, NUM_MediumID_DEN_TrackerMuons, NUM_TightID_DEN_TrackerMuons, NUM_LoosePFIso_DEN_LooseID, NUM_LoosePFIso_DEN_MediumID, NUM_LoosePFIso_DEN_TightID, NUM_TightPFIso_DEN_TightID, NUM_LooseMiniIso_DEN_LooseID, NUM_LooseMiniIso_DEN_MediumID, NUM_IsoMu24_DEN_CutBasedIdMedium_and_PFIsoMedium, NUM_IsoMu24_DEN_CutBasedIdTight_and_PFIsoTight
	Muon ScaRe	corrections/data/MUO/MuonScaRe/Run3-22EFGSep23-Summer22EE-NanoA0Dv12/muon_scalesmearing.json	standard; VXBS payload: corrections/data/MUO/MuonScaRe/Run3-22EFGSep23-Summer22EE-NanoA0Dv12/muon_scalesmearing_VXBS.json
	Muon FSR	corrections/muon_fsr.py	algorithmic; no payload
	Jet veto map	/cvmfs/cms-griddata.cern.ch/cat/metadata/JME/Run3-22EFGSep23-Summer22EE-NanoA0Dv12/latest/jetvetomaps.json.gz	Summer22EE_23Sep2023_RunEFG_V1
	b-tag WP	/cvmfs/cms-griddata.cern.ch/cat/metadata/BTV/Run3-22EFGSep23-Summer22EE-NanoA0Dv12/latest/btagging.json.gz	PNet_wp_values (L, M, T)
Run3_2023	Golden JSON	corrections/data/golden_json/Cert_Collisions2022_355100_362760_Golden.json	data luminosity mask
	Trigger SF	HLT_IsoMu24	NUM_IsoMu24_DEN_CutBasedIdMedium_and_PFIsoMedium
	LHE PDF uncertainty	LHEPdfWeight	replicas_rms: nominal index 0; replicas 1-100; up=1+RMS, down=max(0,1-RMS); alphaS excluded; missing branch=unity
	LHE QCD scales	LHEScaleWeight	muR0p5_muF0p5=index 0; muR0p5_muF1=index 1; muR1_muF0p5=index 3; muR1_muF2=index 5; muR2_muF1=index 7; muR2_muF2=index 8; dataset-normalized six-point prescription; QCD_ren_scale_{process}: muR0p5_muF1 / muR2_muF1; QCD_fac_scale_{process}: muR1_muF0p5 / muR1_muF2; denominator sums are stored in skim reports; missing sums=skip
	JEC/JER	/cvmfs/cms-griddata.cern.ch/cat/metadata/JME/Run3-23CSep23-Summer23-NanoA0Dv12/latest/jet_jerc.json.gz	JEC MC: Summer23Prompt23_V4_MC; JEC data: Summer23Prompt23_Run{}_V4_DATA, Summer23Prompt23_V4_DATA; JER: Summer23Prompt23_RunCv1234_JRV3_MC
	JER smearing	/cvmfs/cms-griddata.cern.ch/cat/metadata/JME/JER-Smearing/latest/jer_smear.json.gz	sources: JER, Total
	Pileup	/cvmfs/cms-griddata.cern.ch/cat/metadata/LUM/Run3-23CSep23-Summer23-NanoA0Dv12/latest/puWeights.json.gz	Collisions2023_366403_369802_eraBC_GoldenJson
	Muon SF	/cvmfs/cms-griddata.cern.ch/cat/metadata/MUO/Run3-23CSep23-Summer23-NanoA0Dv12/latest/muon_Z.json.gz	NUM_LooseID_DEN_TrackerMuons, NUM_MediumID_DEN_TrackerMuons, NUM_TightID_DEN_TrackerMuons, NUM_LoosePFIso_DEN_LooseID, NUM_LoosePFIso_DEN_MediumID, NUM_LoosePFIso_DEN_TightID, NUM_TightPFIso_DEN_TightID, NUM_LooseMiniIso_DEN_LooseID, NUM_LooseMiniIso_DEN_MediumID, NUM_IsoMu24_DEN_CutBasedIdMedium_and_PFIsoMedium, NUM_IsoMu24_DEN_CutBasedIdTight_and_PFIsoTight
	Muon ScaRe	corrections/data/MUO/MuonScaRe/Run3-23CSep23-Summer23-NanoA0Dv12/muon_scalesmearing.json	standard; VXBS payload: corrections/data/MUO/MuonScaRe/Run3-23CSep23-Summer23-NanoA0Dv12/muon_scalesmearing_VXBS.json
	Muon FSR	corrections/muon_fsr.py	algorithmic; no payload
Run3_2023	Jet veto map	/cvmfs/cms-griddata.cern.ch/cat/metadata/JME/Run3-23CSep23-Summer23-NanoA0Dv12/latest/jetvetomaps.json.gz	Summer23Prompt23_RunC_V1
	b-tag WP	/cvmfs/cms-griddata.cern.ch/cat/metadata/BTV/Run3-23CSep23-Summer23-NanoA0Dv12/latest/btagging.json.gz	PNet_wp_values (L, M, T)
	Golden JSON	corrections/data/golden_json/Cert_Collisions2023_366442_370790_Golden.json	data luminosity mask
	Trigger SF	HLT_IsoMu24	NUM_IsoMu24_DEN_CutBasedIdMedium_and_PFIsoMedium
	LHE PDF uncertainty	LHEPdfWeight	replicas_rms: nominal index 0; replicas 1-100; up=1+RMS, down=max(0,1-RMS); alphaS excluded; missing branch=unity

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Table 1 – continued

Era	Correction	Payload / path	Key / tag
Run3_2023BPix	LHE QCD scales	LHEScaleWeight	muR0p5_muF0p5=index 0; muR0p5_muF1=index 1; muR1_muF0p5=index 3; muR1_muF2=index 5; muR2_muF1=index 7; muR2_muF2=index 8; dataset-normalized six-point prescription; QCD_ren_scale_{process}: muR0p5_muF1 / muR2_muF1; QCD_fac_scale_{process}: muR1_muF0p5 / muR1_muF2; denominator sums are stored in skim reports; missing sums=skip
	JEC/JER	/cvms/cms-griddata.cern.ch/cat/metadata/JME/Run3-23DSep23-Summer23BPix-NanoA0Dv12/latest/jet_jerc.json.gz	JEC MC: Summer23BPixPrompt23_V4_MC; JEC data: Summer23BPixPrompt23_Run{ }_V4_DATA, Summer23BPixPrompt23_V4_DATA; JER: Summer23BPixPrompt23_RunD_JRV3_MC
	JER smearing	/cvms/cms-griddata.cern.ch/cat/metadata/JME/JER-Smearing/latest/jer_smear.json.gz	sources: JER, Total
	Pileup	/cvms/cms-griddata.cern.ch/cat/metadata/LUM/Run3-23DSep23-Summer23BPix-NanoA0Dv12/latest/puWeights.json.gz	Collisions2023_369803_370790_eraD_GoldenJson
	Muon SF	/cvms/cms-griddata.cern.ch/cat/metadata/MUO/Run3-23DSep23-Summer23BPix-NanoA0Dv12/latest/muon_Z.json.gz	NUM_LooseID_DEN_TrackerMuons, NUM_MediumID_DEN_TrackerMuons, NUM_TightID_DEN_TrackerMuons, NUM_LoosePFIso_DEN_LooseID, NUM_LoosePFIso_DEN_MediumID, NUM_LoosePFIso_DEN_TightID, NUM_TightPFIso_DEN_TightID, NUM_LooseMiniIso_DEN_LooseID, NUM_LooseMiniIso_DEN_MediumID, NUM_IsoMu24_DEN_CutBasedIdMedium_and_PFIsoMedium, NUM_IsoMu24_DEN_CutBasedIdTight_and_PFIsoTight
	Muon ScaRe	corrections/data/MUO/MuonScaRe/Run3-23DSep23-Summer23BPix-NanoA0Dv12/muon_scalesmearing.json	standard; VXBS payload: corrections/data/MUO/MuonScaRe/Run3-23DSep23-Summer23BPix-NanoA0Dv12/muon_scalesmearing_VXBS.json
	Muon FSR	corrections/muon_fsr.py	algorithmic; no payload
	Jet veto map	/cvms/cms-griddata.cern.ch/cat/metadata/JME/Run3-23DSep23-Summer23BPix-NanoA0Dv12/latest/jetvetomaps.json.gz	Summer23BPixPrompt23_RunD_V1
	b-tag WP	/cvms/cms-griddata.cern.ch/cat/metadata/BTV/Run3-23DSep23-Summer23BPix-NanoA0Dv12/latest/btagging.json.gz	PNet_wp_values (L, M, T)
	Golden JSON	corrections/data/golden_json/Cert_Collisions2023_366442_370790_Golden.json	data luminosity mask
Run3_2024	Trigger SF	HLT_IsoMu24	NUM_IsoMu24_DEN_CutBasedIdMedium_and_PFIsoMedium
	LHE PDF uncertainty	LHEPdfWeight	replicas_rms: nominal index 0; replicas 1-100; up=1+RMS, down=max(0,1-RMS); alphaS excluded; missing branch=unity
	LHE QCD scales	LHEScaleWeight	muR0p5_muF0p5=index 0; muR0p5_muF1=index 1; muR1_muF0p5=index 3; muR1_muF2=index 5; muR2_muF1=index 7; muR2_muF2=index 8; dataset-normalized six-point prescription; QCD_ren_scale_{process}: muR0p5_muF1 / muR2_muF1; QCD_fac_scale_{process}: muR1_muF0p5 / muR1_muF2; denominator sums are stored in skim reports; missing sums=skip
	JEC/JER	/cvms/cms-griddata.cern.ch/cat/metadata/JME/Run3-24CDEReprocessingFGHIPrompt-Summer24-NanoA0Dv15/latest/jet_jerc.json.gz	JEC MC: Summer24Prompt24_V5_MC; JEC data: Summer24Prompt24_V5_DATA; JER: Summer24Prompt24_JRV2_MC
	JER smearing	/cvms/cms-griddata.cern.ch/cat/metadata/JME/JER-Smearing/latest/jer_smear.json.gz	sources: JER, Total
	Pileup	/cvms/cms-griddata.cern.ch/cat/metadata/LUM/Run3-24CDEReprocessingFGHIPrompt-Summer24-NanoA0Dv15/latest/puWeights_BCDEFGHI.json.gz	Collisions24_BCDEFGHI_goldenJSON
	Muon SF	/cvms/cms-griddata.cern.ch/cat/metadata/MUO/Run3-24CDEReprocessingFGHIPrompt-Summer24-NanoA0Dv15/latest/muon_Z.json.gz	NUM_LooseID_DEN_TrackerMuons, NUM_MediumID_DEN_TrackerMuons, NUM_TightID_DEN_TrackerMuons, NUM_LoosePFIso_DEN_LooseID, NUM_LoosePFIso_DEN_MediumID, NUM_LoosePFIso_DEN_TightID, NUM_TightPFIso_DEN_TightID, NUM_IsoMu24_DEN_CutBasedIdMedium_and_PFIsoMedium, NUM_IsoMu24_DEN_CutBasedIdTight_and_PFIsoTight
	Muon ScaRe	corrections/data/MUO/MuonScaRe/Run3-24CDEReprocessingFGHIPrompt-Summer24-NanoA0Dv15/muon_scalesmearing.json	standard; VXBS payload: corrections/data/MUO/MuonScaRe/Run3-24CDEReprocessingFGHIPrompt-Summer24-NanoA0Dv15/muon_scalesmearing_
	Muon FSR	corrections/muon_fsr.py	algorithmic; no payload
	Jet veto map	/cvms/cms-griddata.cern.ch/cat/metadata/JME/Run3-24CDEReprocessingFGHIPrompt-Summer24-NanoA0Dv15/latest/jetvetomaps.json.gz	Summer24Prompt24_RunBCDEFGHI_V1
Run3_2025	b-tag WP	/cvms/cms-griddata.cern.ch/cat/metadata/BTV/Run3-24CDEReprocessingFGHIPrompt-Summer24-NanoA0Dv15/latest/btagging.json.gz	UParTAK4_wp_values (L, M, T)
	Golden JSON	corrections/data/golden_json/Cert_Collisions2024_378981_386951_Golden.json	data luminosity mask
	Trigger SF	HLT_IsoMu24	NUM_IsoMu24_DEN_CutBasedIdMedium_and_PFIsoMedium
	LHE PDF uncertainty	LHEPdfWeight	replicas_rms: nominal index 0; replicas 1-100; up=1+RMS, down=max(0,1-RMS); alphaS excluded; missing branch=unity
	LHE QCD scales	LHEScaleWeight	muR0p5_muF0p5=index 0; muR0p5_muF1=index 1; muR1_muF0p5=index 3; muR1_muF2=index 5; muR2_muF1=index 7; muR2_muF2=index 8; dataset-normalized six-point prescription; QCD_ren_scale_{process}: muR0p5_muF1 / muR2_muF1; QCD_fac_scale_{process}: muR1_muF0p5 / muR1_muF2; denominator sums are stored in skim reports; missing sums=skip
	JEC/JER	/cvms/cms-griddata.cern.ch/cat/metadata/JME/Run3-25Prompt-Summer24-NanoA0Dv15/latest/jet_jerc.json.gz	JEC MC: Summer24Prompt25_V3_MC; JEC data: Summer24Prompt25_V3_DATA; JER: Summer24Prompt25_JRV2_MC
	JER smearing	/cvms/cms-griddata.cern.ch/cat/metadata/JME/JER-Smearing/latest/jer_smear.json.gz	sources: JER, Total
	Pileup	/cvms/cms-griddata.cern.ch/cat/metadata/LUM/Run3-25Prompt-Summer24-NanoA0Dv15/latest/puWeights_2025pp_Golden_Summer24_25ns_69200ub.json.gz	Collisions25_goldenJSON
	Muon SF	/afs/cern.ch/work/v/vdamante/Hmm_newSkim/corrections/data/MUO/SF/Run3-25Prompt-Summer24-NanoA0Dv15/muon_Z.json.gz	NUM_LooseID_DEN_TrackerMuons, NUM_MediumID_DEN_TrackerMuons, NUM_TightID_DEN_TrackerMuons, NUM_LoosePFIso_DEN_LooseID, NUM_LoosePFIso_DEN_MediumID, NUM_LoosePFIso_DEN_TightID, NUM_TightPFIso_DEN_TightID, NUM_LooseMiniIso_DEN_LooseID, NUM_LooseMiniIso_DEN_MediumID, NUM_IsoMu24_DEN_CutBasedIdMedium_and_PFIsoMedium, NUM_IsoMu24_DEN_CutBasedIdTight_and_PFIsoTight
	Muon ScaRe	corrections/data/MUO/MuonScaRe/Run3-25Prompt-Summer24-NanoA0Dv15/muon_scalesmearing.json	standard; VXBS payload: corrections/data/MUO/MuonScaRe/Run3-25Prompt-Summer24-NanoA0Dv15/muon_scalesmearing_VXBS.json
Run3_2025	Muon FSR	corrections/muon_fsr.py	algorithmic; no payload
	Jet veto map	/cvms/cms-griddata.cern.ch/cat/metadata/JME/Run3-25Prompt-Summer24-NanoA0Dv15/latest/jetvetomaps.json.gz	Summer24Prompt25_RunBCDEFG_V1
	b-tag WP	/cvms/cms-griddata.cern.ch/cat/metadata/BTV/Run3-25Prompt-Summer24-NanoA0Dv15/latest/btagging.json.gz	UParTAK4_wp_values (L, M, T)
	Golden JSON	corrections/data/golden_json/Cert_Collisions2025_391658_398903_Golden.json	data luminosity mask
	Trigger SF	HLT_IsoMu24	NUM_IsoMu24_DEN_CutBasedIdMedium_and_PFIsoMedium

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Table 1 – continued

Era	Correction	Payload / path	Key / tag
Run3_2026	LHE PDF uncertainty	LHEPdfWeight	replicas_rms: nominal index 0; replicas 1-100; up=1+RMS, down=max(0,1-RMS); alphaS excluded; missing branch=unity
	LHE QCD scales	LHEScaleWeight	muR0p5_muF0p5=index 0; muR0p5_muF1=index 1; muR1_muF0p5=index 3; muR1_muF2=index 5; muR2_muF1=index 7; muR2_muF2=index 8; dataset-normalized six-point prescription; QCD_ren_scale_{process}: muR0p5_muF1 / muR2_muF1; QCD_fac_scale_{process}: muR1_muF0p5 / muR1_muF2; denominator sums are stored in skim reports; missing sums=skip
	JEC/JER	/cvmfs/cms-griddata.cern.ch/cat/metadata/JME/Run3-26Prompt-Summer24-NanoA0Dv15/latest/jet_jerc.json.gz	JEC MC: Summer24Prompt26_V2_MC; JEC data: Summer24Prompt26_Run{}_V1_DATA, Summer24Prompt26_V1_DATA; JER: Summer24Prompt25_JRV2_MC
	JER smearing	/cvmfs/cms-griddata.cern.ch/cat/metadata/JME/JER-Smearing/latest/jer_smear.json.gz	sources: JER, Total
	Pileup	/cvmfs/cms-griddata.cern.ch/cat/metadata/LUM/Run3-25Prompt-Summer24-NanoA0Dv15/latest/puWeights_2025pp_Golden_Summer24_25ns_69200ub.json.gz	Collisions25_goldenJSON
	Muon SF	/cvmfs/cms-griddata.cern.ch/cat/metadata/MU0/Run3-26Prompt-Summer24-NanoA0Dv15/latest/muon_Z.json.gz	
	Muon ScaRe	corrections/data/MU0/MuonScaRe/Run3-26Prompt-Summer24-NanoA0Dv15/muon_scalesmearing.json	standard; VXBS payload: corrections/data/MU0/MuonScaRe/Run3-26Prompt-Summer24-NanoA0Dv15/muon_scalesmearing_VXBS.json
	Muon FSR	corrections/muon_fsr.py	algorithmic; no payload
	Jet veto map	/cvmfs/cms-griddata.cern.ch/cat/metadata/JME/Run3-26Prompt-Summer24-NanoA0Dv15/latest/jetvetomaps.json.gz	Summer24Prompt26_RunBCD_V1
	b-tag WP	/cvmfs/cms-griddata.cern.ch/cat/metadata/BTV/Run3-26Prompt-Summer24-NanoA0Dv15/latest/btagging.json.gz	UParTAK4_wp_values (L, M, T)
	Golden JSON	corrections/data/golden_json/Cert_Collisions2026_401624_403937_golden.json	data luminosity mask
	Trigger SF	HLT_IsoMu24	NUM_IsoMu24_DEN_CutBasedIdMedium_and_PFIsoMedium
	LHE PDF uncertainty	LHEPdfWeight	replicas_rms: nominal index 0; replicas 1-100; up=1+RMS, down=max(0,1-RMS); alphaS excluded; missing branch=unity
	LHE QCD scales	LHEScaleWeight	muR0p5_muF0p5=index 0; muR0p5_muF1=index 1; muR1_muF0p5=index 3; muR1_muF2=index 5; muR2_muF1=index 7; muR2_muF2=index 8; dataset-normalized six-point prescription; QCD_ren_scale_{process}: muR0p5_muF1 / muR2_muF1; QCD_fac_scale_{process}: muR1_muF0p5 / muR1_muF2; denominator sums are stored in skim reports; missing sums=skip

LHE PDF uncertainty. The nominal member is LHEPdfWeight[0] and replicas 1–100 are used. For each event, the relative replica RMS is $\sigma_{\text{PDF}} = \sqrt{\sum_i (w_i/w_0 - 1)^2 / (N_{\text{valid}} - 1)}$. The multiplicative variations are $w_{\text{PDF}}^{\text{Up}} = 1 + \sigma_{\text{PDF}}$ and $w_{\text{PDF}}^{\text{Down}} = \max(0, 1 - \sigma_{\text{PDF}})$. The configured alphaS members are not included.

LHE QCD-scale uncertainty. μ_R is the QCD renormalization scale, which controls the scale at which α_s is evaluated. μ_F is the factorization scale, which separates long-distance PDF evolution from the short-distance hard process. Values 0.5, 1, and 2 are multiplicative factors relative to the nominal generator scales. The configured final nuisances are: QCD_ren_scale: Down= $(\mu_R, \mu_F) = (0.5, 1)$ (LHEScaleWeight[1]), Up=(2, 1) (index 7); and QCD_fac_scale: Down=(1, 0.5) (index 3), Up=(1, 2) (index 5). The corner points (0.5, 0.5) (index 0) and (2, 2) (index 8) are listed and normalized in the skim metadata but are not used by the two final Up/Down pairs. Each scale template is normalized with its own dataset-wide denominator from the skim reports before histogram filling.